

THE SUMMER EE GLOW-UP

10 resources and exactly how to use each one.
Come back in September dangerous.

```
buildlist $ start glow-up --resources 10 --weeks 12  
loading summer plan ... ready.
```

READ THIS FIRST**Most EE students waste the summer and wonder why they feel behind in September.**

Three months is enough time to go from someone who studies circuits to someone who builds things people do not believe came out of a dorm room. The resources below are the fastest path. But a list is not a plan, so this guide tells you exactly how to use each one and how to stack them into a real summer. Every resource has a link. Tap the name to open it.

THE ONE RULE

Do not try to do all 10. Pick 3 that point at the same job and go deep. Depth is what gets you hired and what makes content nobody can fake. A checklist is not a skill.

THE 10 RESOURCES

Each one with what it is, how to actually use it, and the job it maps to. Links are clickable.

01 Learning the Art of Electronics

BOOK | CORE | Hayes & Horowitz

> open [search for the book](#)

The lab companion to the Art of Electronics bible. A real bench course, not a textbook.

HOW TO USE IT

- > Treat it as a lab. Grab a breadboard, a function generator, and a cheap USB scope.
- > One lab per sitting. Build it, measure it, then read why it behaved that way.
- > Do not skip the transistor and op-amp labs. That is where students fake understanding.

-> **The job this maps to:** Analog hardware engineer. Every EE interview eventually drifts back to this material.

02 Phil's Lab

YOUTUBE and COURSE | INTERMEDIATE | PCB, DSP, STM32

> open youtube.com/@PhilsLab

Real-world PCB design, firmware, and DSP, all built on actual STM32 boards.

HOW TO USE IT

- > Watch the KiCad STM32 series once with no laptop, then again building along.
- > Clone one of his boards end to end, then change one thing. Swap a sensor, add a regulator.
- > Get it fabbed. A board in your hand beats ten tutorials you only watched.

-> **The job this maps to:** Embedded hardware engineer. PCB and firmware on one resume is rare and it pays.

03 Ben Eater

YOUTUBE and KIT | CORE | digital logic

> open eater.net

Build an 8-bit CPU on breadboards, or the 6502 kit. The deepest understanding on this list.

HOW TO USE IT

- > Build the breadboard CPU or the 6502 kit. Do not just watch the videos.
- > Film it. The clock module alone is stop-the-scroll content for the feed.
- > When it breaks, debug with a multimeter before you watch his fix.

-> **The job this maps to:** Digital design and computer architecture. Proves you understand machines from the gate up.

04 Contextual Electronics

COURSE | INTERMEDIATE | Chris Gammell

> open contextualelectronics.com

A structured KiCad bootcamp where the output is a working device you keep.

HOW TO USE IT

- > Follow it as your KiCad sprint. The point is a finished board, not slides.
- > Pair it with your NextPCB account and actually order the thing.
- > Post the build log weekly. Documenting it is the second product.

-> **The job this maps to:** Hardware engineer who can take a product from schematic to a fabricated board.

05 MIT OpenCourseWare

COURSE | CORE | 6.002 Circuits, 6.302 Feedback

> open [ocw.mit.edu 6.002](https://ocw.mit.edu/6.002)

Free and rigorous. The theory backbone under everything else you build this summer.

HOW TO USE IT

- > One lecture, one problem set, repeat. Do not binge lectures or it will not stick.
- > 6.302 Feedback is the secret weapon. Control is where self-taught builders stop.
- > Keep a notebook of every equation you actually used in a build. That is your real syllabus.

-> **The job this maps to:** Any rigorous EE role. Proof you can do the math, not just wire things together.

06 nandland and HDLBits

SITE | ADVANCED | Verilog and FPGA

> open nandland.com | hdlbits.01xz.net

Learn hardware description and digital design, then drill it until it feels boring.

HOW TO USE IT

- > Learn Verilog basics on nandland, then grind HDLBits problems to muscle memory.
- > Buy a cheap FPGA board like a Tang Nano or iCE40 and blink an LED in real hardware.
- > Build one real thing. A UART, a VGA pattern, or a tiny CPU.

-> **The job this maps to:** FPGA and digital design engineer. Almost no student builder touches this, so you stand out.

07 GNU Radio and an RTL-SDR

TOOL | INTERMEDIATE | RF and DSP

> open gnuradio.org | rtl-sdr.com

A 25 dollar dongle that turns the airwaves into a playground. On-brand for your sky content.

HOW TO USE IT

- > Decode FM first, then ADS-B aircraft, then NOAA weather satellite images.
- > Build a flowgraph from scratch instead of loading someone else's.
- > This feeds your satellite and eavesdrop-on-the-sky episodes directly.

-> **The job this maps to:** RF and DSP engineer. Capturing real signals is a portfolio piece, not a class assignment.

08 Making Embedded Systems

BOOK and PODCAST | INTERMEDIATE | Elecia White

> open embedded.fm

The book that gets you past Arduino into real firmware. Pair it with the Embedded.fm podcast.

HOW TO USE IT

- > Read it next to an STM32 dev board. Apply each chapter to code that same night.
- > Touch registers, interrupts, and a real debugger. No more guessing.
- > Use the podcast on commutes to pick up how working firmware engineers actually talk.

-> **The job this maps to:** Firmware engineer. The single most hireable skill on this entire list.

09 Falstad and ngspice

TOOL | CORE | simulation

> open falstad.com | ngspice.sourceforge.io

Falstad builds intuition fast. ngspice gives you the real numbers when you need them.

HOW TO USE IT

- > Use Falstad to feel circuits. Drag one out, watch the current animate, change a value.
- > Move to ngspice when you need SPICE models and numbers you can trust.
- > Study what makes Falstad feel good. You are building CircuitIQ to beat exactly this.

-> **The job this maps to:** Anyone doing circuit design. Simulating before you solder saves money and ego.

10 EEVblog, The Amp Hour, r/AskElectronics

COMMUNITY | CORE | the unstuck button

> open eevblog.com | theamphour.com

Where working engineers and serious hobbyists hang out. Use it when a build stalls.

HOW TO USE IT

- > Stuck for over an hour? Post it. Photos, what you tried, the actual measurement.
- > Lurk The Amp Hour podcast to absorb how experienced engineers reason.
- > Answer one beginner question a week. Teaching is how you find your own gaps.

-> **The job this maps to:** Not a job, a multiplier. People who ask good questions get unstuck and get referred.

THE 12-WEEK STACK

How to combine them so the summer actually compounds. Each phase ends with something you ship.

PHASE	DO THIS	YOU SHIP
FOUNDATION Weeks 1-4	Learning the Art of Electronics labs next to MIT 6.002. Falstad open every day.	An analog circuit you built and can explain cold.
BUILD Weeks 5-8	Contextual Electronics or Phil's Lab. Design and fab one board. Start firmware with Making Embedded Systems.	A real PCB running your own code.
SPIKE Weeks 9-12	Pick one lane and go far. FPGA, RF, or deep firmware. Stop spreading yourself thin.	One project nobody believes a student built, plus the build log.

PICK YOUR LANE

If you only remember one thing from this guide, remember this. Three resources, one direction.

THE HARDWARE LANE

You want to design and fab boards.

- > Learning the Art of Electronics
- > Phil's Lab
- > Contextual Electronics

Outcome: a real PCB running your firmware, in your hand by September.

THE SIGNALS LANE

You want to understand and capture signals.

- > MIT 6.002 and 6.302
- > GNU Radio and RTL-SDR
- > Falstad and ngspice

Outcome: you pull real signals out of the air and explain the math behind them.

THE DIGITAL LANE

You want to build computers and the code on them.

- > Ben Eater
- > nandland and HDLBits
- > Making Embedded Systems

Outcome: you build a machine from logic up and write the firmware that drives it.

THE GLOW-UP IS NOT THIS PDF

This guide is a map, not a trophy. The glow-up is the boards on your bench in September, not the file in your downloads. Pick your lane today. Start phase one this week.

KEEP THE MOMENTUM

> THE BUILD LIST

The newsletter. Weekly build breakdowns and the projects you should not be able to make at home.

> FOLLOW THE BUILD

@1nska on Instagram. Daily build logs and the projects behind every guide.

Build in public. Ship by September.

Follow @1nska if you're building the future, not just studying for it.